

BORON FAMILY

SECTION 1 — NCERT STATEMENT QUESTIONS (Q1–Q16)

Q1. Occurrence, abundance and electronic configuration**Answer:** 13 statements are correct.**Correct statements:** 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13.**Incorrect statements:** 14.**Why the incorrect statements are wrong**14. Incorrect — Kernite and borax do not have the same hydration. Correct statement: Kernite is $\text{Na}_2\text{B}_4\text{O}_7 \cdot 4\text{H}_2\text{O}$, whereas borax is $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$.**Q2. Periodic anomalies and their causes****Answer:** 14 statements are correct.**Correct statements:** 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.**Incorrect statements:** 15.**Why the incorrect statements are wrong**

15. Incorrect — Filled 3d electrons shield the nuclear charge poorly, not more effectively than s/p electrons. Correct statement: Poor 3d shielding increases effective nuclear attraction and makes Ga slightly smaller than Al.

Q3. Master periodic-order audit — Group 13**Answer:** 18 statements are correct.**Correct statements:** 1, 2, 3, 4, 5, 7, 8, 9, 10, 12, 13, 14, 15, 16, 17, 18, 19, 20.**Incorrect statements:** 6, 11.**Why the incorrect statements are wrong**6. Incorrect — The third ionisation enthalpy order printed here is wrong. Correct statement: Correct IE_3 order: $\text{In} < \text{Al} < \text{Tl} < \text{Ga} < \text{B}$.11. Incorrect — The relative positions of In and Tl in the melting-point order are reversed. Correct statement: Correct melting-point order: $\text{Ga} < \text{In} < \text{Tl} < \text{Al} < \text{B}$.**Q4. Chemical-trend logic****Answer:** 14 statements are correct.**Correct statements:** 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.**Incorrect statements:** 15.**Why the incorrect statements are wrong**15. Incorrect — The inert pair is the ns_2 electron pair, not the np_1 electron. Correct statement: Down Group 13, the ns_2 pair increasingly remains non-bonding; the np_1 electron participates in the +1 state.**Q5. Borax: identity, structure, hydrolysis and heating****Answer:** 12 statements are correct.**Correct statements:** 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12.**Incorrect statements:** 13, 14.**Why the incorrect statements are wrong**13. Incorrect — Borax solution is alkaline, not acidic. Correct statement: Borax hydrolyses in water: $\text{Na}_2\text{B}_4\text{O}_7 + 7\text{H}_2\text{O} \rightarrow 2\text{NaOH} + 4\text{H}_3\text{BO}_3$.14. Incorrect — Not all ten waters are external lattice water in the structural representation. Correct statement: Borax is $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$: four OH groups are structural and eight H_2O molecules are lattice water.**Q6. Borax bead test****Answer:** 12 statements are correct.**Correct statements:** 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 13, 14.**Incorrect statements:** 11, 12.**Why the incorrect statements are wrong**11. Incorrect — The cobalt bead is not produced by reduction of CoO to cobalt metal. Correct statement: $\text{CoO} + \text{B}_2\text{O}_3 \rightarrow \text{Co}(\text{BO}_2)_2$, giving a blue cobalt(II) metaborate bead.12. Incorrect — B_2O_3 is not inert toward the metal oxides used in the bead test. Correct statement: Molten B_2O_3 acts as a glass-forming acidic oxide/solvent and reacts with metal oxides to form coloured borates/metaborates.**Q7. Orthoboric acid: preparation, properties and structure****Answer:** 11 statements are correct.**Correct statements:** 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11.**Incorrect statements:** 12, 13, 14.**Why the incorrect statements are wrong**12. Incorrect — Solid boric acid is not built only from tetrahedral BO_4 units. Correct statement: H_3BO_3 contains planar BO_3 units joined by hydrogen bonding into layers.13. Incorrect — Boric acid is not a strong mineral acid. Correct statement: H_3BO_3 is a weak monobasic Lewis acid.

14. Incorrect — Its solubility does not decrease on heating. Correct statement: Boric acid is sparingly soluble in cold water and much more soluble in hot water.

Q8. Orthoboric acid: acidity, heating and use**Answer:** 12 statements are correct.**Correct statements:** 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13.**Incorrect statements:** 10, 14.**Why the incorrect statements are wrong**10. Incorrect — The three O-H hydrogens do not ionise independently in three normal steps. Correct statement: Boric acid is monobasic; it accepts OH^- from water to form $[\text{B}(\text{OH})_4]^-$.

14. Incorrect — Boric acid is not a strong reducing agent in water. Correct statement: Its characteristic acid behaviour is Lewis acidity, not reducing action.

Q9. Diborane: preparation, properties, combustion and hydrolysis**Answer:** 15 statements are correct.**Correct statements:** 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 16.**Incorrect statements:** 15.**Why the incorrect statements are wrong**15. Incorrect — Hydration of B_2O_3 gives boric acid; it does not produce diborane. Correct statement: $\text{B}_2\text{O}_3 + 3\text{H}_2\text{O} \rightarrow 2\text{H}_3\text{BO}_3$.**Q10. Diborane: Lewis-base cleavage, ammonia and borazine chemistry****Answer:** 11 statements are correct.**Correct statements:** 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 14.**Incorrect statements:** 11, 12, 13.**Why the incorrect statements are wrong**11. Incorrect — CO can donate an electron pair to an electron-deficient BH_3 fragment. Correct statement: $\text{B}_2\text{H}_6 + 2\text{CO} \rightarrow 2\text{BH}_3 \cdot \text{CO}$.12. Incorrect — Borazine is not a hydrocarbon. Correct statement: Borazine is $\text{B}_3\text{N}_3\text{H}_6$, a B-N ring compound called inorganic benzene.

13. Incorrect — Ammonia does not convert diborane directly and exclusively to BN. Correct statement: It first gives an ammonia-diborane ionic adduct; heating can form borazine, and stronger heating can ultimately give BN.

Q11. Diborane: structure and bonding**Answer:** 11 statements are correct.**Correct statements:** 1, 2, 3, 4, 5, 6, 7, 8, 9, 13, 14.**Incorrect statements:** 10, 11, 12.

Why the incorrect statements are wrong

10. Incorrect — The four terminal H atoms and two bridging H atoms have different environments. Correct statement: Diborane contains four terminal H and two bridging H; they are not all equivalent.

11. Incorrect — Diborane does not contain a conventional B=B double bond. Correct statement: Its electron deficiency is accommodated by two B-H-B three-centre two-electron bridges.

12. Incorrect — The bridge bonds are not ordinary 2c-2e B-H bonds. Correct statement: Terminal B-H bonds are 2c-2e; each B-H-B bridge is a 3c-2e bond.

Q12. Tetrahydridoborates / borohydrides

Answer: 11 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 14.

Incorrect statements: 11, 12, 13.

Why the incorrect statements are wrong

11. Incorrect — $[\text{BH}_4]^-$ is not square planar. Correct statement: $[\text{BH}_4]^-$ is tetrahedral.

12. Incorrect — NaBH_4 does not act as an oxidising agent. Correct statement: NaBH_4 is a reducing agent and hydride donor.

13. Incorrect — Hydrolysis of diborane does not prepare borohydrides. Correct statement: $\text{LiBH}_4/\text{NaBH}_4$ can be prepared from the corresponding metal hydride and diborane in ether: $2\text{MH} + \text{B}_2\text{H}_6 \rightarrow 2\text{MBH}_4$.

Q13. Uses of boron, borax, boric acid and borides

Answer: 14 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

Incorrect statements: None.

Q14. Aluminium: properties, alloys and uses

Answer: 15 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 16.

Incorrect statements: 15.

Why the incorrect statements are wrong

15. Incorrect — Aluminium is not avoided because it cannot form useful alloys. Correct statement: Aluminium forms many useful alloys, including duralumin and magnalium.

Q15. General reactions and reactivity

Answer: 13 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13.

Incorrect statements: 14.

Why the incorrect statements are wrong

14. Incorrect — Concentrated HNO_3 does not attack aluminium rapidly under ordinary conditions. Correct statement: Concentrated HNO_3 passivates aluminium by maintaining/strengthening its protective oxide film.

Q16. Assertion-reason logic

Answer: 13 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13.

Incorrect statements: 14.

Why the incorrect statements are wrong

14. Incorrect — The assertion B_2O_3 is acidic is true, but the reason is false: boron is not the most metallic Group 13 element. Correct statement: Boron is the least metallic member of Group 13; B_2O_3 is acidic because boron is strongly non-metallic/covalent.

SECTION 2 — SINGLE CORRECT (Q17–Q31)**Q17.**

Answer: Option 2 — statements 1, 2, 3, 4.

Correct statements: 1, 2, 3, 4.

Incorrect statements: 5, 6.

Why the incorrect statements are wrong

5. Incorrect — Diborane cleavage is driven by donor-acceptor interaction at electron-deficient boron. Correct statement: Lewis bases donate to BH_3 fragments during cleavage.

6. Incorrect — Different donors do not necessarily give the same cleavage product. Correct statement: Small amines can give unsymmetrical cleavage, while bulky/tertiary donors can give neutral $\text{BH}_3 \cdot \text{L}$ adducts.

Q18.

Answer: Option 4 — statements 1, 2, 3, 6.

Correct statements: 1, 2, 3, 6.

Incorrect statements: 4, 5.

Why the incorrect statements are wrong

4. Incorrect — The central and terminal iodine atoms in I_3^- do not have identical electronic environments. Correct statement: I_3^- is linear; the two terminal I atoms are equivalent to each other, while the central I is different.

5. Incorrect — TlI_3 is not best described as $\text{Tl}^{3+} + 3\text{I}^-$. Correct statement: It is classically formulated as $\text{Tl}^+(\text{I}_3^-)$, consistent with the stability of Tl(I) .

Q19.

Answer: Option 1 — statements 1, 2, 3, 4.

Correct statements: 1, 2, 3, 4.

Incorrect statements: 5, 6.

Why the incorrect statements are wrong

5. Incorrect — Borazine and benzene do not have identical reactivity toward HCl . Correct statement: Borazine is more reactive toward addition because B-N bonds are polar.

6. Incorrect — Their charge distributions are not identical. Correct statement: Borazine has polar $\text{B}\delta^+-\text{N}\delta^-$ bonds; benzene has equivalent C-C bonds and much less bond polarity.

Q20.

Answer: Option 3 — statements 1, 2, 3, 4.

Correct statements: 1, 2, 3, 4.

Incorrect statements: 5, 6.

Why the incorrect statements are wrong

5. Incorrect — BeO and Al_2O_3 are not strongly basic oxides only. Correct statement: Both are amphoteric.

6. Incorrect — Hydration and hydrolysis strongly affect Be/Al aqueous chemistry. Correct statement: Small highly charged cations are strongly hydrated and their aqua ions can hydrolyse.

Q21.

Answer: Option 1 — statements 1, 2, 3, 4, 5.

Correct statements: 1, 2, 3, 4, 5.

Incorrect statements: 6.

Why the incorrect statements are wrong

6. Incorrect — The hydrated tetraborate ion is not represented with four terminal B=O double bonds. Correct statement: The standard localised structure contains B-O-B bridges and B-OH groups, with no conventional terminal B=O bonds.

Q22.

Answer: Option 4 — statements 1, 2, 3, 4, 6.

Correct statements: 1, 2, 3, 4, 6.

Incorrect statements: 5.

Why the incorrect statements are wrong

5. Incorrect — Diborane bonding is not identical to ordinary ethane bonding. Correct statement: Diborane has four terminal 2c-2e B-H bonds and two electron-deficient 3c-2e B-H-B bridges.

Q23.

Answer: Option 1 — statements 1, 2, 3, 5, 6.

Correct statements: 1, 2, 3, 5, 6.

Incorrect statements: 4.

Why the incorrect statements are wrong

4. Incorrect — BF_4^- is not an electron-deficient Lewis acid. Correct statement: BF_4^- is a tetrahedral, octet-complete anion and is not the Lewis-acid centre in this set.

Q24.

Answer: Option 3 — statements 1, 2, 3, 4, 6.

Correct statements: 1, 2, 3, 4, 6.

Incorrect statements: 5.

Why the incorrect statements are wrong

5. Incorrect — Fluoride is extremely difficult to oxidise to F_2 ; acidified MnO_2 does not do this under ordinary laboratory conditions. Correct statement: NaF remains fluoride under these conditions; with Ca^{2+} it can give white CaF_2 .

Q25.

Answer: Option 2 — statements 1, 2, 3, 4, 6.

Correct statements: 1, 2, 3, 4, 6.

Incorrect statements: 5.

Why the incorrect statements are wrong

5. Incorrect — Boric acid is not a strong tribasic acid. Correct statement: It is a weak monobasic Lewis acid that accepts OH^- to form $[\text{B}(\text{OH})_4]^-$.

Q26.

Answer: Option 4 — statements 1, 2, 4, 5.

Correct statements: 1, 2, 4, 5.

Incorrect statements: 3, 6.

Why the incorrect statements are wrong

3. Incorrect — Borax is not a strong mineral acid in water. Correct statement: Borax solution is alkaline because the borate ion hydrolyses.

6. Incorrect — B_2O_3 is not the only stable boron species in dilute aqueous borax. Correct statement: Aqueous borax is described by borate/boric-acid equilibria involving $\text{B}(\text{OH})_3$ and $[\text{B}(\text{OH})_4]^-$.

Q27.

Answer: Option 1 — statements 1, 2, 3, 4.

Correct statements: 1, 2, 3, 4.

Incorrect statements: 5, 6.

Why the incorrect statements are wrong

5. Incorrect — $\text{Tl}(\text{I})$ is not less stable than $\text{Tl}(\text{III})$ in ordinary aqueous chemistry. Correct statement: The inert-pair effect makes $\text{Tl}(\text{I})$ more stable than $\text{Tl}(\text{III})$.

6. Incorrect — $\text{Tl}(\text{III})$ compounds do exist. Correct statement: $\text{Tl}(\text{III})$ compounds exist but are relatively oxidising because reduction to $\text{Tl}(\text{I})$ is favourable.

Q28.

Answer: Option 3 — statements 1, 2, 3, 4, 5.

Correct statements: 1, 2, 3, 4, 5.

Incorrect statements: 6.

Why the incorrect statements are wrong

6. Incorrect — The practical note does not universally recommend storing water overnight in bare aluminium utensils. Correct statement: Aluminium can react under alkaline/chemical conditions and its surface chemistry depends on the oxide film; such a blanket recommendation is not justified.

Q29.

Answer: Option 2 — statements 1, 2, 3, 4, 5.

Correct statements: 1, 2, 3, 4, 5.

Incorrect statements: 6.

Why the incorrect statements are wrong

6. Incorrect — The hydration enthalpy of Al^{3+} is not too small to matter. Correct statement: Al^{3+} has very high hydration enthalpy, strongly stabilising hydrated species such as $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$.

Q30.

Answer: Option 4 — statements 1, 2, 3, 5.

Correct statements: 1, 2, 3, 5.

Incorrect statements: 4, 6.

Why the incorrect statements are wrong

4. Incorrect — Borax is not a necessary first dehydration product of pure boric acid. Correct statement: Heating H_3BO_3 gives HBO_2 , then condensed boric acids, and finally B_2O_3 .

6. Incorrect — The dehydration sequence does not show that boric acid is a strong tribasic proton donor. Correct statement: Boric acid remains a weak monobasic Lewis acid.

Key audit: The supplied answer key marks Option 4 but later calls statement 5 false. Those two parts conflict. Chemically, statement 5 (' B_2O_3 can be rehydrated to boric acid under suitable conditions') is acceptable, so Option 4 is retained.

Q31.

Answer: Option 3 — statements 1, 2, 3, 4, 6.

Correct statements: 1, 2, 3, 4, 6.

Incorrect statements: 5.

Why the incorrect statements are wrong

5. Incorrect — Al_4C_3 does not characteristically give acetylene on hydrolysis. Correct statement: Al_4C_3 is a methanide carbide and gives CH_4 : $\text{Al}_4\text{C}_3 + 12\text{H}_2\text{O} \rightarrow 4\text{Al}(\text{OH})_3 + 3\text{CH}_4$.

SECTION 3 — ONE OR MORE CORRECT (Q32–Q37)

Q32.

Answer: Select statements 1, 2, 3, 4.

Correct statements: 1, 2, 3, 4.

Incorrect statements: 5, 6.

Why the incorrect statements are wrong

5. Incorrect — The two bridging H atoms are different from the four terminal H atoms. Correct statement: Diborane contains four terminal H and two bridging H.

6. Incorrect — The classical bridge-bond model uses sp_3 , not sp_2 , hybridisation at boron. Correct statement: Each B is described as sp_3 hybridised in the elementary model.

Q33.

Answer: Select statements 1, 2, 3, 4, 5.

Correct statements: 1, 2, 3, 4, 5.

Incorrect statements: 6.

Why the incorrect statements are wrong

6. Incorrect — $\text{Al}_2(\text{CH}_3)_6$ is not an ionic assembly of Al^{3+} and CH_3^- ions. Correct statement: It is a covalent, electron-deficient bridged dimer with terminal and bridging methyl groups.

Q34.

Answer: Select statements 1, 2, 3, 4, 5.

Correct statements: 1, 2, 3, 4, 5.

Incorrect statements: 6.

Why the incorrect statements are wrong

6. Incorrect — $\text{Al}(\text{OH})_3$ is not strongly basic and non-amphoteric. Correct statement: $\text{Al}(\text{OH})_3$ is amphoteric: it dissolves in acids and in strong alkali.

Q35.

Answer: Select statements 1, 2, 3, 4, 5.

Correct statements: 1, 2, 3, 4, 5.

Incorrect statements: 6.

Why the incorrect statements are wrong

6. Incorrect — BaCl_2 does not selectively precipitate Cr(III) from untreated chrome alum as BaCrO_4 . Correct statement: Ba^{2+} precipitates sulfate as BaSO_4 ; chromate precipitation requires Cr(VI) /chromate, not ordinary Cr(III) alum solution.

Q36.

Answer: Select statements 1, 2, 3, 4, 5.

Correct statements: 1, 2, 3, 4, 5.

Incorrect statements: 6.

Why the incorrect statements are wrong

6. Incorrect — Solid sodium metaborate need not consist only of isolated monomeric BO_2^- ions. Correct statement: Solid borates commonly contain polymeric B-O frameworks; the simple BO_2^- formula is a stoichiometric representation.

Q37.

Answer: Select statements 1, 2, 3, 4.

Correct statements: 1, 2, 3, 4.

Incorrect statements: 5, 6.

Why the incorrect statements are wrong

5. Incorrect — Two acids can still react by Lewis acid-base/complex formation. Correct statement: Boric acid can form chelated boron complexes with salicylic acid/polyhydroxy ligands.

6. Incorrect — Chelation does not force boron to remain three-coordinate. Correct statement: Chelation can make boron four-coordinate, approximately tetrahedral.

SECTION 4 — MATCH THE COLUMNS (Q38–Q47)**Q38.**

Match: A–S, B–Q, C–R, D–P.

Reason: Diborane hydrolysis gives H_3BO_3 (W). Heating H_3BO_3 gives HBO_2 , then $\text{H}_2\text{B}_4\text{O}_7$, and strongest heating gives B_2O_3 .

Q39.

Match: A–Q; B–P; C–R,S; D–Q.

Reason: $\text{BBr}_3 + \text{H}_2$ is reduction; the borax/Cu system is the bead test; moist AlCl_3 gives white HCl fumes by hydrolysis; $\text{Cr}_2\text{O}_3 + \text{Al}$ is aluminothermic reduction.

Q40.

Match: A–P, B–Q, C–R, D–S.

Reason: BF_3 is trigonal-planar/electron-deficient; solid AlCl_3 is polymeric/high-coordinate; Al_2Cl_6 has two Cl bridges; B_2H_6 has 3c-2e B-H-B bridges.

Q41.

Chemically complete match: A–P,S; B–Q,S; C–Q,S; D–R,S.

Bank-key convention: A–P,S; B–Q,S; C–Q,S; D–R.

Why: $\text{NH}_3/\text{MeNH}_2/\text{Me}_2\text{NH}$ are Lewis-base donors and are placed in the unsymmetrical-cleavage class. Me_3N and CO give neutral $\text{BH}_3 \cdot \text{L}$ adducts. NaH supplies H^- to make $[\text{BH}_4]^-$; because H^- is itself an electron-pair donor, D–S is also chemically valid if S is read literally.

Audit note: The matrix wording says “every chemically appropriate match”, so omitting D–S makes the printed key under-complete. If the examiner intends S to mean only neutral donor-ligand interaction, then the bank-key convention is the intended answer.

Q42.

Match: A–P, B–Q, C–R, D–S.

Reason: Borazine = inorganic benzene; h-BN = graphite-like layered solid; B_4C = hard abrasive; Al_4C_3 = methanide carbide giving CH_4 on hydrolysis.

Q43.

Match: A–P,R,S; B–P,Q,S; C–P,Q,S; D–P,R,S.

Reason: This follows the classifications used in the supplied question bank for Lewis acidity, dimerisation, delocalisation and hydrolysis.

Q44.

Match: Set I: A–S, B–T, C–R, D–P. Set II: A–S, B–R, C–P, D–Q.

Reason: Borax on strong heating gives $\text{NaBO}_2 + \text{B}_2\text{O}_3$; B_2H_6 hydrolysis gives H_3BO_3 ; $\text{B}_2\text{H}_6/\text{NH}_3$ on strong heating gives BN; hydrolysed SiCl_4 on heating gives SiO_2 ; $\text{BCl}_3 + \text{LiAlH}_4$ gives B_2H_6 .

Q45.

Verified Set I: A–P,Q,S; B–R,T; C–P; D–P,Q.

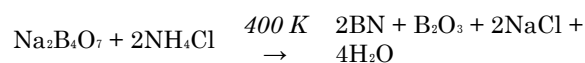
Set II: A–Q; B–R; C–S; D–P.

Why Set I:

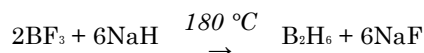
A. Borazine hydrolysis gives boric acid, ammonia and hydrogen. Therefore a gaseous product is formed (P), boric acid is monobasic (Q), and NH_3 blackens moistened mercurous-nitrate paper (S).



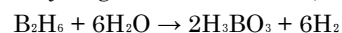
B. Borax + NH_4Cl on heating gives BN and B_2O_3 ; BN is graphite-like and B_2O_3 is an acidic oxide. Hence B–R,T.



C. $\text{BF}_3 + \text{NaH}$ gives diborane and NaF. Diborane is a gas, so C–P. NaF neither blackens mercurous-nitrate paper nor is the oxygen-bleach/brightener sodium peroxoborate; therefore the uploaded key's C–S match is chemically incorrect.



D. Diborane hydrolysis gives H_2 and H_3BO_3 ; therefore D–P,Q.

**Q46.**

Match: A–Q,R,T; B–P,S,T; C–Q,R,S,T; D–P,R,S,T.

Reason: B_2H_6 and BF_3 are gases; H_3BO_3 and BI_3 are solids; hydrolysis of boron halides/hydrides gives boron oxyacid products; the listed boron compounds are covalent and trigonal boron occurs in $\text{H}_3\text{BO}_3/\text{BX}_3$.

Q47.

Match: Set I: A–Q,S; B–P; C–S; D–Q. Set II: A–R; B–P; C–S; D–T; E–Q.

Reason: These are the matches used by the supplied bank: borax contains sp_2/sp_3 B; potash alum gives the K^+ cobaltinitrite test; borazole has sp_2 B; and the common names/uses are AlCl_3 —Friedel-Crafts, alum—antiseptic, ultramarine—blue silicate, borax—tincal, magnalium—alloy.

SECTION 5 — COUNT THE CORRECT STATEMENTS (Q48–Q54)**Q48.**

Answer: 10 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9, 12.

Incorrect statements: 10, 11.

Why the incorrect statements are wrong

10. Incorrect — The tetraborate ion does not have all four boron centres tetrahedral. Correct statement: It contains two trigonal and two tetrahedral boron centres.

11. Incorrect — Borax does not contain a conventional B=B pi bond. Correct statement: Its framework is made of B-O bonds, including B-O-B bridges.

Q49.

Answer: 10 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

Incorrect statements: 11, 12.

Why the incorrect statements are wrong

11. Incorrect — Not every B-H contact is an ordinary 2c-2e bond. Correct statement: Only the four terminal B-H bonds are 2c-2e; the two bridges are 3c-2e.

12. Incorrect — Bridge B-H contacts are not shorter and stronger than terminal B-H bonds. Correct statement: Bridge B-H contacts are longer and weaker than terminal B-H bonds.

Q50.

Answer: 10 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 9, 10, 12.

Incorrect statements: 8, 11.

Why the incorrect statements are wrong

8. Incorrect — $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ is not the same electron-deficient species as dry monomeric AlCl_3 . Correct statement: Hydration produces strongly solvated/hydrolysed aluminium(III) species with a different bonding description.

11. Incorrect — Hydration does not destabilise Al^{3+} . Correct statement: The very large hydration enthalpy strongly stabilises hydrated Al^{3+} in water.

Q51.

Answer: 10 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

Incorrect statements: 11, 12.

Why the incorrect statements are wrong

11. Incorrect — Borazine and benzene do not have identical reactivity toward HCl. Correct statement: Borazine is more susceptible to addition because its B-N bonds are polar.

12. Incorrect — B-N bonds are not non-polar. Correct statement: Boron and nitrogen have different electronegativities, giving $\text{B}\delta^+ - \text{N}\delta^-$ bond polarity.

Q52.

Answer: 10 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9, 11.

Incorrect statements: 10, 12.

Why the incorrect statements are wrong

10. Incorrect — Boric acid is not a strong tribasic protonic acid. Correct statement: It is a weak monobasic Lewis acid.

12. Incorrect — Boric acid can react with salicylic acid by complex formation. Correct statement: Salicylic acid can form a four-coordinate chelated boron complex.

Q53.

Answer: 11 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11.

Incorrect statements: 12.

Why the incorrect statements are wrong

12. Incorrect — Calcium is not the preferred clean reducing metal in the cited B_2O_3 -to-boron scheme. Correct statement: Magnesium is commonly used: $\text{B}_2\text{O}_3 + 3\text{Mg} \rightarrow 2\text{B} + 3\text{MgO}$, followed by acid washing.

Q54.

Answer: 3 statements are correct.

Correct statements: 1, 2, 4.

Incorrect statements: 3, 5, 6.

Why the incorrect statements are wrong

3. Incorrect — One mole of borax does not require only one mole of a monobasic strong acid for full conversion to boric acid. Correct statement: $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O} + 2\text{HCl} \rightarrow 4\text{H}_3\text{BO}_3 + 2\text{NaCl} + 5\text{H}_2\text{O}$; thus 2 mol H^+ are required per mol borax.

5. Incorrect — Boric acid can react/dissolve in hydrofluoric acid through fluoroborate chemistry. Correct statement: With excess HF: $\text{H}_3\text{BO}_3 + 4\text{HF} \rightarrow \text{HBF}_4 + 3\text{H}_2\text{O}$.

6. Incorrect — At sufficiently strong heating, the B-N-H system does not stop at borazine as the final product. Correct

statement: Borazine can form on heating the diborane-ammonia adduct, but stronger heating can ultimately give BN.

SECTION 6 — NUMERICAL RESPONSE (Q55–Q74)**Q55.**

Answer: 6.

Working: The two B atoms and four terminal H atoms are coplanar; the two bridge H atoms lie above and below that plane.

Q56.

Answer: 4.

Working: Diborane has four terminal B-H bonds; each is a normal 2c-2e bond.

Q57.

Answer: 2.

Working: There are two B-H-B bridges, each a 3c-2e bond.

Q58.

Answer: 12.

Working: Valence electrons = $2(3) + 6(1) = 12$.

Q59.

Answer: 2.

Working: $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$ contains two trigonal (sp^2) and two tetrahedral (sp^3) B centres.

Q60.

Answer: 2.

Working: Two of the four B centres are tetrahedral.

Q61.

Answer: 5.

Working: The standard tetraborate framework contains five B-O-B bridging oxygen atoms.

Q62.

Answer: 14.

Working: Five bridging O atoms give 10 B-O sigma bonds, plus four terminal B-O(H) bonds: $10 + 4 = 14$.

Q63.

Answer: 8.

Working: $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$ contains eight lattice-water molecules.

Q64.

Answer: 6.

Working: All three B and all three N ring atoms in borazine are sp^2 hybridised.

Q65.

Answer: 12.

Working: The six ring atoms and six directly bonded H atoms are coplanar in the ideal planar borazine model.

Q66.

Answer: 1.5 mol H_2 .

Working: $\text{B}_2\text{H}_6 + 6\text{H}_2\text{O} \rightarrow 2\text{H}_3\text{BO}_3 + 6\text{H}_2$. Therefore $0.25 \times 6 = 1.5$ mol.

Q67.

Answer: 6.0 mol $\text{B}(\text{OCH}_3)_3$.

Working: 1 mol B_2H_6 gives 2 mol $\text{B}(\text{OCH}_3)_3$; 3.0 mol gives 6.0 mol.

Q68.

Answer: 2.0 mol H_3BO_3 .

Working: 1 mol $\text{Na}_2\text{B}_4\text{O}_7$ gives 4 mol H_3BO_3 ; $0.50 \times 4 = 2.0$ mol.

Q69.

Answer: 2.0 mol B_2O_3 .

Working: $2\text{H}_3\text{BO}_3 \rightarrow \text{B}_2\text{O}_3 + 3\text{H}_2\text{O}$; 4.0 mol H_3BO_3 gives 2.0 mol B_2O_3 .

Q70.**Answer:** 3.0 mol H₂.**Working:** 2Al gives 3H₂ in the NaOH/H₂O reaction; 2.0 mol Al therefore gives 3.0 mol H₂.**Q71.****Answer:** 6.0 mol total gas.**Working:** 2NaBH₄ gives 1 mol B₂H₆ + 1 mol H₂ = 2 mol gas. Thus 6 mol NaBH₄ gives 6 mol gaseous products.**Q72.****Answer:** 6 statements are correct.**Correct statements:** (i), (ii), (iii), (iv), (v), (vii).**Incorrect statements:** (vi), (viii).(vi). Incorrect — BI₃ does not have the strongest pπ–pπ overlap. Correct statement: pπ–pπ donation/overlap is strongest for B–F and decreases from BF₃ to BI₃.(viii). Incorrect — BF₃ is not a stronger Lewis acid than BI₃ in the familiar BX₃ order. Correct statement: Lewis acidity increases BF₃ < BCl₃ < BBr₃ < BI₃.**Q73.****Answer:** 3.**Working:** Beryl is Be₃Al₂Si₆O₁₈, so n = 3.**Q74.****Answer:** 2.**Working:** Al₂Cl₆ contains two bridging chlorides.**SECTION 7 — INTEGRATED QUESTIONS (Q75–Q85)****Q75.****Answer:** Option 3 — statements 1, 2, 3, 4.**Correct statements:** 1, 2, 3, 4.**Incorrect statements:** 5, 6.**Why the incorrect statements are wrong**5. Incorrect — CO is a donor to BH₃ but is not classified as an unsymmetrical amine-type cleavage reagent. Correct statement: CO forms a donor adduct such as BH₃·CO.6. Incorrect — Primary, secondary and tertiary amines do not necessarily give identical products. Correct statement: Small amines can give unsymmetrical cleavage; tertiary/bulky donors can give neutral BH₃·L adducts.**Q76.****Answer:** Option 2 — statements 1, 2, 3, 4.**Correct statements:** 1, 2, 3, 4.**Incorrect statements:** 5, 6.**Why the incorrect statements are wrong**

5. Incorrect — Peroxoborate is not only a reducing agent. Correct statement: It is an active-oxygen oxidant and can oxidise Cr(III) to Cr(VI) in alkaline medium.

6. Incorrect — Its oxidation chemistry is directly related to peroxide/active oxygen. Correct statement: The peroxy O–O linkages are the origin of its oxidising/bleaching action.

Q77.**Answer:** Option 4 — statements 1, 2, 3, 4.**Correct statements:** 1, 2, 3, 4.**Incorrect statements:** 5, 6.**Why the incorrect statements are wrong**5. Incorrect — Hydrolysis of BCl₃ does not produce HF. Correct statement: BCl₃ + 3H₂O → H₃BO₃ + 3HCl.6. Incorrect — Silica is not inert toward aqueous HF. Correct statement: HF attacks SiO₂/glass, forming fluorosilicate species.**Q78.****Answer:** Option 1 — statements 1, 2, 3, 4, 6.**Correct statements:** 1, 2, 3, 4, 6.**Incorrect statements:** 5.**Why the incorrect statements are wrong**5. Incorrect — Ca²⁺ is not a transition-metal ion and does not give the most diagnostic coloured borax bead. Correct statement: The diagnostic colours arise from transition-metal borate/metaborate species.**Q79.****Answer:** Option 3 — statements 1, 2, 3, 4, 6.**Correct statements:** 1, 2, 3, 4, 6.**Incorrect statements:** 5.**Why the incorrect statements are wrong**5. Incorrect — Water does not directly reduce borax to elemental boron. Correct statement: A standard route is borax → H₃BO₃ → B₂O₃, followed by reduction with Mg/another suitable reducing agent.**Q80.****Answer:** Statements 1, 2, 3 and 4 are correct.**Correct statements:** 1, 2, 3, 4.**Incorrect statements:** 5, 6.**Why the incorrect statements are wrong**5. Incorrect — Aqueous AlCl₃ electrolysis does not industrially produce aluminium; water is reduced preferentially. Correct statement: Hall-Heroult electrolysis uses Al₂O₃ dissolved in molten cryolite.6. Incorrect — Aluminium is not liberated at the anode. Correct statement: Al³⁺ is reduced to Al at the cathode; carbon is oxidised at the anode.**Q81.****Answer:** 5 statements are correct.**Correct statements:** (i), (ii), (iii), (iv), (v).**Incorrect statements:** (vi).

(vi). Incorrect — The bank does not recommend keeping water overnight in bare aluminium utensils as a universal practice. Correct statement: Aluminium is protected by an oxide film, but its surface can be attacked under suitable chemical/alkaline conditions; no blanket recommendation should be made.

Q82.**Answer:** 7 statements are correct.**Correct statements:** (i), (ii), (iii), (iv), (v), (vi), (vii).**Incorrect statements:** (viii).

(viii). Incorrect — A diagonal relationship means similarities, not identical formulae for every hydride and halide. Correct statement: B/Si and Be/Al show important similarities, but their compounds are not formula-for-formula identical.

Q83.**Answer:** 13 statements are correct.**Correct statements:** 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13.**Incorrect statements:** 14, 15.**Why the incorrect statements are wrong**14. Incorrect — Group 13 halides are not all monomeric trigonal molecules in every state. Correct statement: For example, AlCl₃ forms Al₂Cl₆/associated structures and some heavier halides are ionic or mixed-valence.

15. Incorrect — All six H atoms in diborane are not equivalent terminal 2c-2e hydrogens. Correct statement: There are four terminal 2c-2e H atoms and two bridging 3c-2e H atoms.

Q84.(a) **Answer:** D is the incorrect reaction.**Why D is incorrect:** BCl₃ normally forms a 1:1 Lewis adduct with pyridine; the neutral 1:2 product Cl₃B(py)₂ is not the standard product.**Correct reaction:** BCl₃ + C₅H₅N → BCl₃·C₅H₅N.

(b) Answer: 0 statements are scientifically correct.

A. Incorrect — Not all boron trihalides are simply 'hydrolysed completely' under every condition. Correct statement: Hydrolysis behaviour depends on the halide and medium; BF_3 can give fluoroborate/ HBF_4 -containing species.

B. Incorrect — Anhydrous AlCl_3 is not ionic. Correct statement: Anhydrous AlCl_3 is predominantly covalent and can associate as Al_2Cl_6 .

C. Incorrect — The blanket claim that AlCl_3 is weaker than BCl_3 is not correct in the intended Lewis-acid comparison. Correct statement: Lewis acidity depends on the donor/medium; AlCl_3 is a strong Lewis acid and should not be universally ranked below BCl_3 .

D. Incorrect — Each Al in molecular Al_2Cl_6 is not six-coordinate. Correct statement: Each Al is coordinated to four Cl atoms: two terminal and two bridging.

Q85.

(a) A: $[\text{BH}_2(\text{NH}_3)_2]^+[\text{BH}_4]^-$.

B: $\text{B}_3\text{N}_3\text{H}_6$ (borazine).

C: H_2 .

Atomic number of the element in C: 1 (hydrogen).

NH_3 molecules in the cationic part of A: 2.

Key sequence: $\text{B}_2\text{H}_6 + 2\text{NH}_3 \rightarrow [\text{BH}_2(\text{NH}_3)_2]^+[\text{BH}_4]^-$; on heating, the ammonia-borane system eliminates H_2 and forms borazine.

(b) Lewis acids: 3 — H^+ , B_2H_6 , CH_3^+ .

Lewis bases: 9 — benzene, H_2O , CH_3^- , $(\text{C}_2\text{H}_5)_2\text{O}$, PH_3 , acetone, H^- , CO , NO .

SECTION 8 — EXTENDED STATEMENT COUNT (Q86–Q90)

Q86.

Answer: 9 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9.

Incorrect statements: 10, 11, 12.

Why the incorrect statements are wrong

10. Incorrect — The term peroxo is not merely nominal. Correct statement: Sodium peroxoborate contains genuine O-O peroxide linkages.

11. Incorrect — Sodium metaborate does not contain the same peroxide linkage. Correct statement: NaBO_2 is a borate/metaborate; the O-O linkage is characteristic of peroxoborate.

12. Incorrect — Peroxoborate is not only a reducing agent. Correct statement: It is an active-oxygen oxidising/bleaching agent.

Q87.

Answer: 9 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 12.

Incorrect statements: 9, 10, 11.

Why the incorrect statements are wrong

9. Incorrect — Elemental boron does not give an intense crimson atomic flame like lithium. Correct statement: The diagnostic boron flame tests use volatile boron compounds such as BF_3 or borate esters, which give green coloration.

10. Incorrect — The stated HF-mediated glass attack requires moisture/hydrolysis. Correct statement: Perfectly dry BF_3 does not rapidly etch silica by an aqueous-HF mechanism.

11. Incorrect — Trimethyl borate is not non-volatile. Correct statement: $\text{B}(\text{OCH}_3)_3$ is volatile, which is why it is useful in the green-flame test.

Q88.

Answer: 10 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

Incorrect statements: 11, 12.

Why the incorrect statements are wrong

11. Incorrect — Borazine is not more aromatically stable than benzene. Correct statement: Its polar B-N bonds give weaker aromatic stabilisation and make it more reactive toward addition.

12. Incorrect — Borazine and benzene are not equally resistant to hydrolysis. Correct statement: Borazine hydrolyses much more readily than benzene.

Q89.

Answer: 10 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9, 12.

Incorrect statements: 10, 11.

Why the incorrect statements are wrong

10. Incorrect — GaCl_2 is not a genuine simple Ga(II) paramagnetic compound in this description. Correct statement: It is formulated as mixed-valence $\text{Ga}^+[\text{GaCl}_4]^-$.

11. Incorrect — TlI_3 is not TI^{3+} with three independent iodide ions. Correct statement: It is best represented as $\text{TI}^+(\text{I}_3^-)$, with linear triiodide.

Q90.

Answer: 9 statements are correct.

Correct statements: 1, 2, 3, 4, 5, 6, 7, 8, 9.

Incorrect statements: 10, 11, 12.

Why the incorrect statements are wrong

10. Incorrect — Anodising does not make the aluminium article the cathode in molten cryolite. Correct statement: The aluminium article is the anode in an acidic electrolyte such as dilute H_2SO_4 .

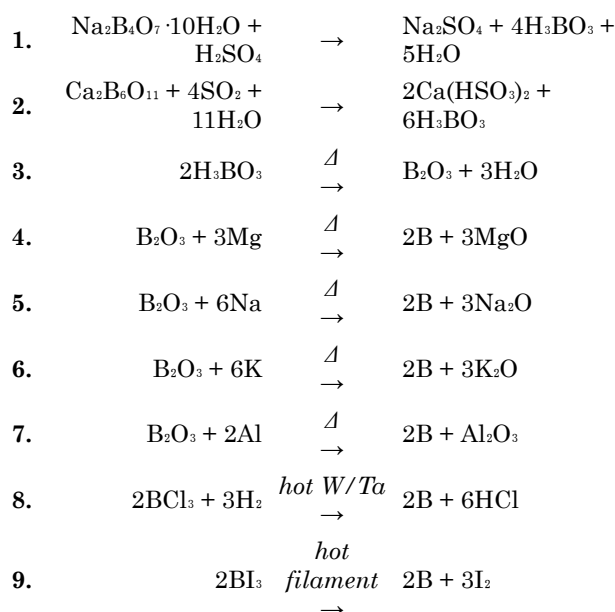
11. Incorrect — Thermite does not work because Al_2O_3 has a less negative formation enthalpy. Correct statement: Thermite is driven by the very favourable formation of Al_2O_3 ; its formation enthalpy is very negative.

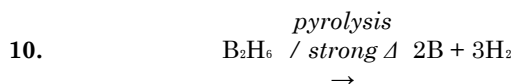
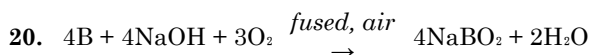
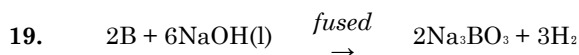
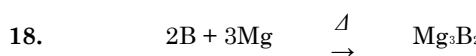
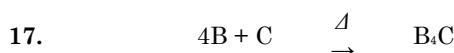
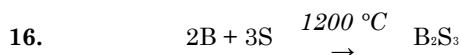
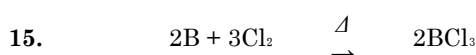
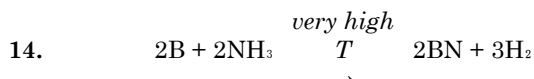
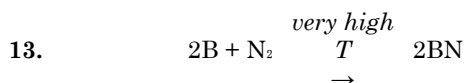
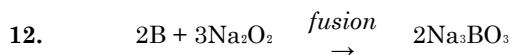
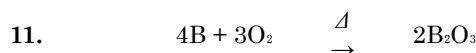
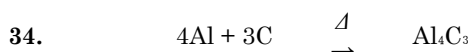
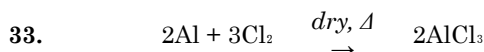
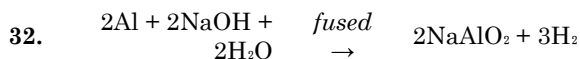
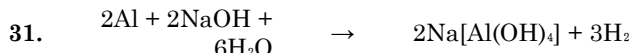
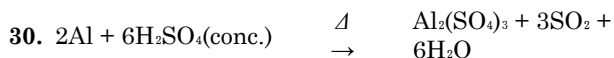
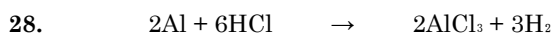
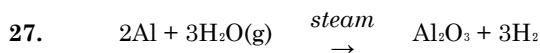
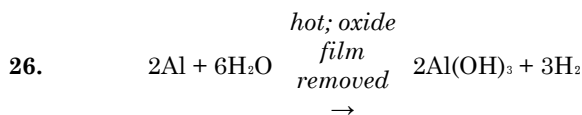
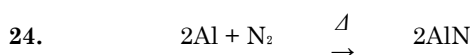
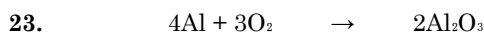
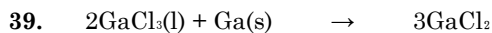
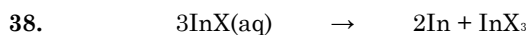
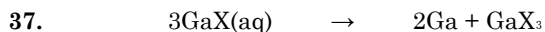
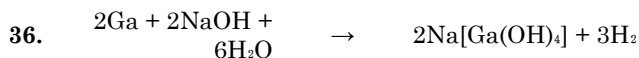
12. Incorrect — Neutral AlCl_3 is not simply recovered unchanged in Friedel-Crafts acylation. Correct statement: AlCl_3 forms complexes/ AlCl_4^- and often binds strongly to the acylated product, so stoichiometric work-up may be required.

SECTION 9 — COMPLETE THE REACTION: VERIFIED TEXTBOOK ANSWER KEY (1–155)

Reaction convention: reactants on the left, reaction condition above the arrow, products on the right. All equations are atom-balanced unless the original item itself is explicitly identified as defective.

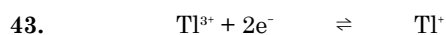
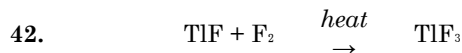
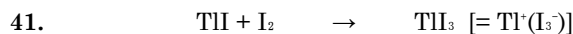
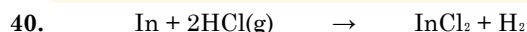
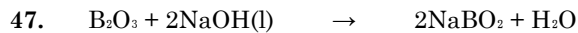
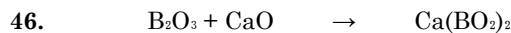
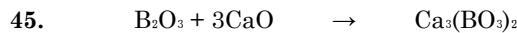
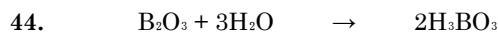
A. Elemental boron — preparation



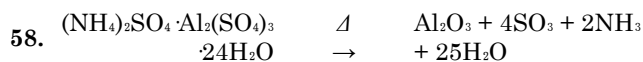
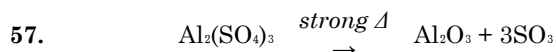
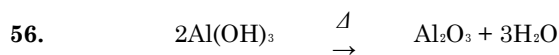
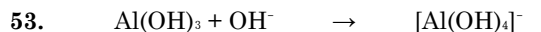
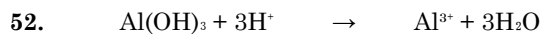
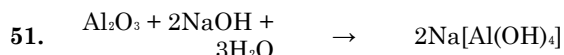
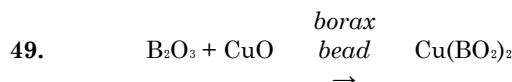
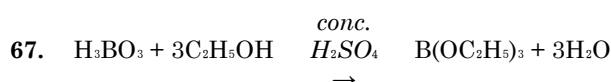
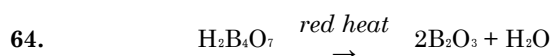
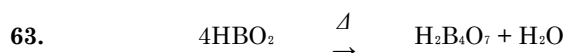
**B. Elemental boron — reactions****C. Aluminium — reactions****D. Gallium, indium, thallium**

Audit note: The uploaded key gives an unbalanced shorthand “ $\text{GaCl}_3 + \text{Ga} \rightarrow 2\text{GaCl}_2$ ”. The balanced empirical equation is shown. “ GaCl_2 ” is a mixed-valence lower chloride, often represented

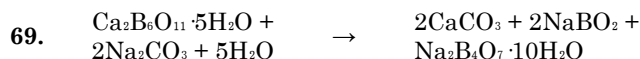
through Ga(I)/Ga(III) units rather than simple Ga(II).

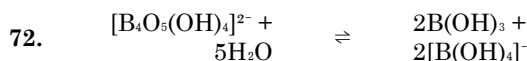
**E. Oxides and hydroxides**

Audit note: Correction to the previous draft: B_2O_3 can behave as a basic oxide toward the very strong acidic oxide P_2O_5 , forming boron phosphate BPO_4 .

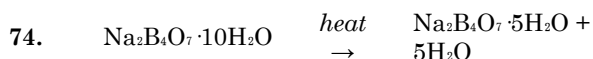
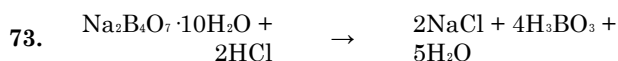
**F. Boric acid, H_3BO_3** 

Audit note: The printed reactant coefficient is $2\text{H}_3\text{BO}_3$; with that stoichiometry the balanced hot-carbonate product is metaborate as shown. If tetraborate is intended, use: $4\text{H}_3\text{BO}_3 + \text{Na}_2\text{CO}_3 \rightarrow \text{Na}_2\text{B}_4\text{O}_7 + \text{CO}_2 + 6\text{H}_2\text{O}$.

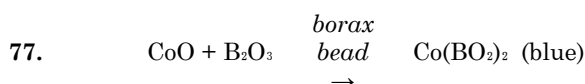
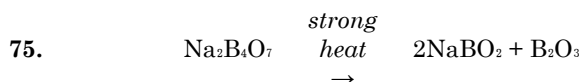
G. Borax and borates



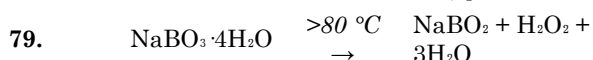
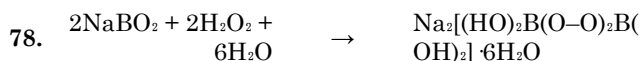
Audit note: This is the buffer-form hydrolysis of the structural tetraborate ion: boric acid plus tetrahydroxyborate are formed together.



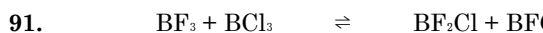
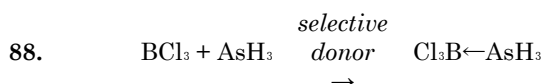
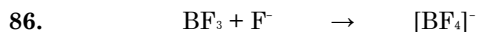
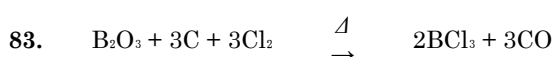
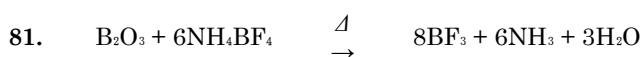
Audit note: The question explicitly asks the first dehydration step. Further heating ultimately gives anhydrous $\text{Na}_2\text{B}_4\text{O}_7$ and then the $\text{NaBO}_2/\text{B}_2\text{O}_3$ borax bead on strong heating.



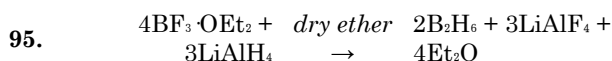
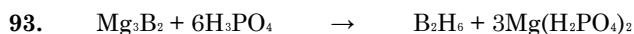
H. Sodium peroxoborate (perborate)



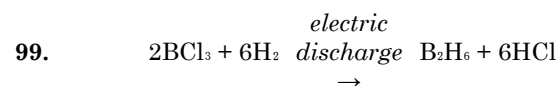
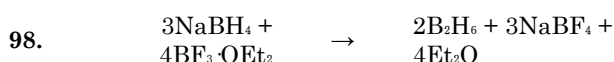
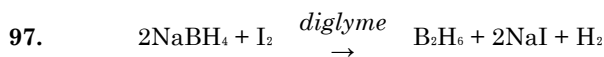
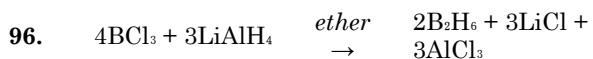
I. Boron trihalides and fluoroboric acid



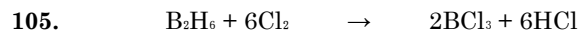
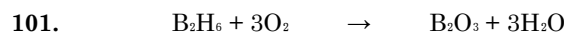
J. Diborane — preparation



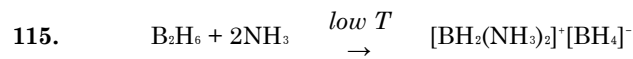
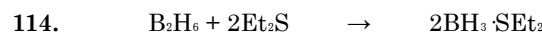
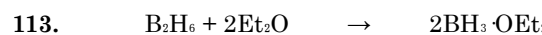
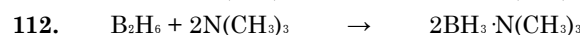
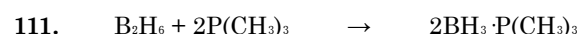
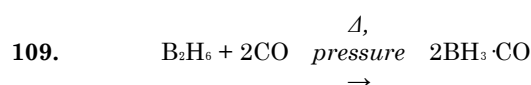
Audit note: The uploaded reaction key writes the fluoride-containing inorganic product as LiAlF_4 . Writing $\text{LiF} + \text{AlF}_3$ is an alternative bookkeeping form, but the source-key product is retained here.



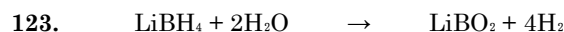
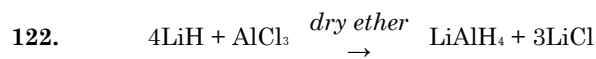
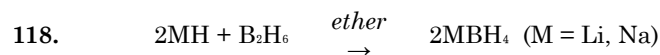
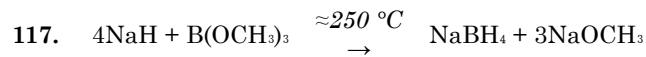
K. Diborane — reactions and cleavage



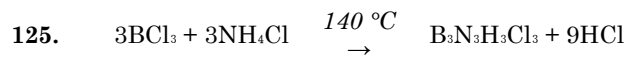
Audit note: Source-key coefficient error: the uploaded key prints 3Cl_2 but also gives $2\text{BCl}_3 + 6\text{HCl}$. Complete chlorination requires 6Cl_2 for atom balance.



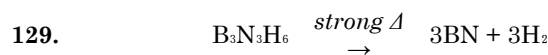
L. Borohydrides and complex hydrides

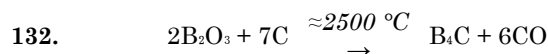
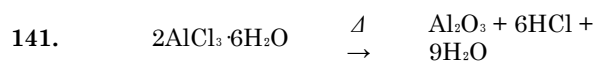
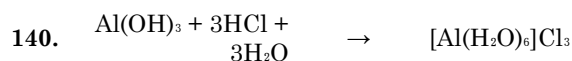
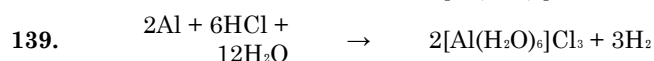
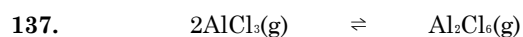


M. Borazine and boron nitride

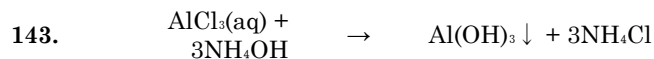


Audit note: The uploaded key leaves the hydride by-products unspecified. A verified balanced form is $\text{B}_3\text{N}_3\text{H}_3\text{Cl}_3 + 3\text{NaBH}_4 \rightarrow \text{B}_3\text{N}_3\text{H}_6 + 3\text{NaCl} + 3/2 \text{B}_2\text{H}_6$.

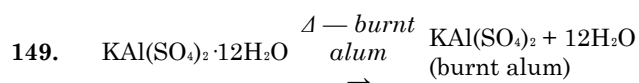
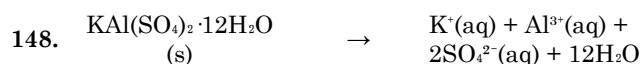


N. Boron carbide, B₄C**O. Aluminothermy (thermite / Goldschmidt)****P. Aluminium chloride and aqueous aluminium chemistry**

Audit note: Use the neutral ammine-adduct notation $\text{AlCl}_3 \cdot 6\text{NH}_3$ for this source reaction; it should not be silently converted into a different aqueous coordination formulation.



Audit note: Source-key / molecular form: $\text{NaAlO}_2 + 2\text{H}_2\text{O}$. In aqueous strong alkali the same amphoteric chemistry is commonly written ionically as $\text{Al}(\text{OH})_3 + \text{OH}^- \rightarrow [\text{Al}(\text{OH})_4]^-$.

**Q. Aluminium sulfate and alums****R. Aluminium extraction — Hall–Héroult / Bayer**